



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 11-Aug-2021

Revision Number 2

## 1. IDENTIFICATION

### Product identifier

**Product Name** IEF Sample Buffer

### Other means of identification

**Catalogue Number(s)** 1610763, 1610763EDU

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### **Technical Service**

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

Acute aquatic toxicity

Category 3

### GHS Label elements, including precautionary statements

#### **Hazard statements**

Harmful to aquatic life

#### **Precautionary Statements - Prevention**

Avoid release to the environment

#### **Precautionary Statements - Response**

#### **Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

No information available

**3. COMPOSITION/INFORMATION ON INGREDIENTS****Substance**

Not applicable

**Mixture**

| Chemical name                 | CAS No  | Weight-% |
|-------------------------------|---------|----------|
| 1,2,3-Propanetriol<br>56-81-5 | 56-81-5 | 50       |

**4. FIRST AID MEASURES****Description of necessary first aid measures**

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Ingestion    | Rinse mouth thoroughly with water.                                                                                   |

**For emergency responders**

**Self-protection of the first aider** No information available.

**Most important symptoms and effects, both acute and delayed**

**Symptoms** No information available.

**Indication of immediate medical attention and special treatment needed, if necessary**

**Note to physicians** Treat symptomatically.

**5. FIREFIGHTING MEASURES****Suitable Extinguishing Media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** No information available.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. ACCIDENTAL RELEASE MEASURES

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. HANDLING AND STORAGE

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control parameters

Exposure guidelines

| Chemical name                 | ACGIH TLV         | OSHA PEL                                                                                                                                                                                                                               | Ontario                   | European Union       |        |
|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|--------|
| 1,2,3-Propanetriol<br>56-81-5 | No data available | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction | -                         | -                    |        |
| Chemical name                 | China             | Japan Society of Occupational Health                                                                                                                                                                                                   | Korea                     | Australia            | Taiwan |
| 1,2,3-Propanetriol<br>56-81-5 | -                 | -                                                                                                                                                                                                                                      | TWA: 10 mg/m <sup>3</sup> | 10 mg/m <sup>3</sup> | -      |

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are

exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                              |                   |                          |                          |
|----------------------------------------------|-------------------|--------------------------|--------------------------|
| Physical state                               | Liquid            | Odour                    | Odourless                |
| Appearance                                   | aqueous solution  | Odour threshold          | No information available |
| Colour                                       | colourless        |                          |                          |
| Property                                     | Values            | Remarks • Method         |                          |
| pH                                           |                   | No information available |                          |
| Melting point / freezing point               |                   | No information available |                          |
| Boiling point / boiling range                | > 100 °C / 212 °F |                          |                          |
| Flash point                                  | 160 °C / 320 °F   |                          |                          |
| Evaporation rate                             |                   | No information available |                          |
| Flammability (solid, gas)                    |                   | No information available |                          |
| Upper/lower flammability or explosive limits |                   |                          |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |                          |
| Lower flammability or explosive limits       | Not applicable    |                          |                          |
| Vapour pressure                              |                   | No information available |                          |
| Vapour density                               |                   | No information available |                          |
| Relative density                             |                   | No information available |                          |
| Solubility(ies)                              |                   |                          |                          |
| Water solubility                             | Miscible in water |                          |                          |
| Solubility in other solvents                 |                   | No information available |                          |
| Partition coefficient                        |                   | No information available |                          |
| Autoignition temperature                     |                   | No information available |                          |
| Decomposition temperature                    |                   | No information available |                          |
| Viscosity                                    |                   |                          |                          |
| Kinematic viscosity                          |                   | No information available |                          |
| Dynamic viscosity                            |                   |                          |                          |

### Other information

**Oxidising properties** Not applicable

## 10. STABILITY AND REACTIVITY

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None

**Sensitivity to static discharge** None.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**11. TOXICOLOGICAL INFORMATION****Information on the likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

**Acute toxicity****Numerical measures of toxicity****Component Information**

| Chemical name      | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|--------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** No information available.

**12. ECOLOGICAL INFORMATION**

**Toxicity**

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

**Ecotoxicity** Harmful to aquatic life.

| Chemical name      | Algae/aquatic plants | Fish                                                 | Crustacea                                   |
|--------------------|----------------------|------------------------------------------------------|---------------------------------------------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> ) | EC50: >500mg/L (24h, <i>Daphnia magna</i> ) |

**Persistence and degradability**

No information available.

**Bioaccumulative potential**

There is no data for this product.

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| 1,2,3-Propanetriol | -1.76                 |

**Other adverse effects**

No information available.

## 13. DISPOSAL CONSIDERATIONS

**Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. TRANSPORT INFORMATION

**IMDG** Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

**IATA** Not regulated

**RID** Not regulated

**ADR** Not regulated

**ADN** Not regulated

**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

## 15. REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. OTHER INFORMATION**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 11-Aug-2021

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**